

time and assembly errors. Carefully assemble the components and double-check for any overlooked error. Connect the assembled circuit to the COM port available at the back side of a computer. The information stored in the microcontroller is sent to the PC through the COM port using the HyperTerminal program.

To open the HyperTerminal program, go to Start → Programs → Accessories → Communications → HyperTerminal. You will see a window as shown in Fig. 5. Type in the desired name and click 'ok'. Select a COM port (refer Fig. 6) while ignoring the other

options and click 'ok'.

Now select the baud rate as '9600', data bits as '8', parity as 'none', stop bits as '1' and flow control as 'none', and click 'ok' (refer Fig. 7).

When you press switches S1 through S8, the microcontroller sends the stored message and you will see the message appear on the screen as shown in Fig. 8.

Software

The software for this project is given at the end of this article. It is written in 'C' language and compiled using Keil µVision3 compiler. The finally ob-

tained '.hex' file is downloaded to the microcontroller using a suitable programmer. The source program is well commented and easy to understand.

In the software, first enable timer and serial interrupts. Then configure the timer by mode and reload value to give the correct speed (9600 bits per second). Next, make the transmit interrupt 1 (TI) move the data to SCON. SCON will give the necessary UART signal and your data will be sent serially.

EFY note. The source code of this project has been included in this month's EFY-CD.

SERIAL.C

```
Serial.C
#include <REGX51.H>
#include <stdio.h>
sbit MES1 = P2^0;
sbit MES2 = P2^1;
sbit MES3 = P2^2;
sbit MES4 = P2^3;
sbit MES5 = P2^4;
sbit MES6 = P2^5;
sbit MES7 = P2^6;
sbit MES8 = P2^7;

int i;
void timelms();
void delay(int n);
void main(void) {
    /* execution starts here after stack init */
    SCON = 0x52; /* SCON:
mode 1, 8-bit UART, enable rcvr */
    TMOD = 0x20; /* TMOD:
timer 1, mode 2, 8-bit reload */
    TH1 = 0x0fd; /* TH1: reload
value for 9600 baud at 11.0592MHz */
    TR1 = 1; /*
TR1: timer 1 run */
    TI = 1; /* TI:
set TI to send first char of UART */
    while (1) {
        P2=0xFF;
        if (MES1 == 0)
        {
            printf ("Welcome to EFY \n
");
            printf ("Microcontroller Based
Serial Data transfer to PC \n");
            printf ("by T.Shankar &
S.Thangamani \n");
            printf ("Data transmission suc-
cessful\n\n");
        }
        delay(1000);
        if (MES2 == 0)
        {
            printf ("T.Shankar &
S.Thangamani\n");
            printf ("Old door NO. 8, Power
House Road, Madurai\n");
            printf ("Contact.- 91-
9443474124, 91-9944082936\n");
            printf ("shankartbe@yahoo.
co.in, parvatyeie@yahoo.co.in\n");
            printf ("Data transmission suc-
cessful \n\n");
            delay(1000);
            if (MES3 == 0)
            {
                printf ("Welcome to EFY \n
");
                printf ("EFY Enterprises Pvt.
Ltd.\n");
                printf ("DS7/1, Phase-I\n");
                printf ("Okhla Industrial Area,
New Delhi-110020\n");
                printf ("Data transmission suc-
cessful \n\n");
                delay(1000);
                if (MES4 == 0)
                {
                    printf ("Tested By Sunil Kumar\
n");
                    printf ("Published in EFY Maga-
zine, Feb.-2010\n\n");
                    delay(1000);
                    if (MES5 == 0)
                    {
                        printf ("Welcome to EFY \n
");
                        printf ("Message-5\n\n");
                        delay(1000);
                        if (MES6 == 0)
                        {
                            printf ("Message-6\n\n");
                            delay(1000);
                            if (MES7 == 0)
                            {
                                printf ("Welcome to EFY \n
");
                                printf ("Message-7\n\n");
                                delay(1000);
                                if (MES8 == 0)
                                {
                                    printf ("Welcome to EFY \n
");
                                    printf ("Message-8\n\n");
                                    delay(1000);
                                }
                            }
                        }
                    }
                }
            }
        }
    }
}

void timelms() /* 1 ms delay */
{
    int i;
    for(i=0;i<50;i++); // appropriate lms
}

void delay(int n) /* do nothing
n*lms */
{
    int i;
    for(i=0;i<n;i++)
        timelms();
}
```

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